EE1007
Introduction to Computer Science Laboratory

Lab 8
Array

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Lab 8.1
Average Score

- You are asked to write a program to calculate the average of three scores and find the fail ones.
- This program will ask the user to enter three integer for the scores. The range of score is [0, 100].
- After that, the program will print the average score with three decimal places.
- If there are score(s) less than 60 points, please display the word “fail:”, following by the order and the failed score(s) in the sequence.
- You need to use an array to store the scores and calculate the average of the scores. Otherwise, you will get 0 point.

Lab 8.1
Average Score

- The screen dialog should appear as follows:

```
Please enter the scores of three students:
30
50
80
avg = 53.333
fail:
1: 30
2: 50

Process returned 0 (0x0)   execution time : 9.100 s
Press any key to continue.
```


Lab 8.2
Arithmetic Sequence

- You are asked to write a program to show the Arithmetic Sequence
- This program will ask the user to enter the three integer for the first term(*a1*), common difference(*d*) and the number of terms(*n*).
- The range of the integer is $[-2^{31}-1, 2^{31}-1]$.
- The program will display the arithmetic series as a1+a2+…+an = sum.
- You must use array to store all terms.
- If the term is an negative integer, it should include parentheses in the above equation as (-a1)+(-a2)+…(-an) = (-sum).

Lab 8.2
Arithmetic Sequence

- The screen dialog should appear as follows:

```
Please enter the first term(a1), the common difference(d) and the number of terms(n):
1
4
5
The arithmetical series: 1 + 5 + 9 + 13 + 17 = 45
Process returned 0 (0x0)   execution time : 6.968 s
Press any key to continue.
```


Lab 8.3
Pascal's Triangle

- You are asked to write a program to print Pascal's Triangle based on a given integer
- This program will ask the user to enter a positive integer n. After that, the program will display Pascal's triangle with *n* columns.
- Then the program will ask the user to enter a positive integer again, and print the corresponding Pascal's triangle with *n* columns .
- The program will terminate until the user enters 0.
- The range if n is [1,30]
- You need to store the values of Pascal's triangle with a multidimension array.

Lab 8.3
Pascal's Triangle

- The screen dialog should appear as follows:

```
Please enter a number for the column of the Pascal triangle:
4
1
1  1
1  2  1
1  3  3  1

Please enter a number for the column of the Pascal triangle:
5
1
1  1
1  2  1
1  3  3  1
1  4  6  4  1

Please enter a number for the column of the Pascal triangle:
8
1
1  1
1  2  1
1  3  3  1
1  4  6  4  1
1  5  10  10  5  1
1  6  15  20  15  6  1
1  7  21  35  35  21  7  1

Please enter a number for the column of the Pascal triangle:
0

Process returned 0 (0x0)   execution time : 18.095 s
Press any key to continue.
```


Lab 8.4
NCU Beverage Ordering System

- The catering of NCU is quite inconvenient and it introduces a big trouble for students.
- To solve this problem, a new company, NCU Beverage, is established at 2021/11/25 for beverage ordering system.
- The company will provide an intelligent beverage ordering system. However, the employees of NCU Beverage do not know how to develop the system and ask for your help.
- In this Lab, you are asked to develop the intelligent beverage ordering system for NCU Beverage company.

Lab 8.4
NCU Beverage Ordering System

- As a start-up company, the NCU Beverage only has three types of beverage in the ordering system: cold drink, coffee and hot drink, respectively.
- To provide rich choices, both cold drink and coffee consist of **eight** different kinds of taste, and hot drink consists of **four** different kinds of taste.



Lab 8.4
NCU Beverage Ordering System

- The program will ask the user to input which kind of drink he/she wants. 1 for cold drink, 2 for coffee and 3 for hot drink.
- After the user inputs which kind of drink he/she wants, the program will ask the user to input which kind of taste he/she wants. (EX:1~4, 1~8)
- And the program will execute from the first action again until the user input 0 to end the system.
- Finally, the program will show all the drink and the price the user has bought.



Lab 8.4
NCU Beverage Ordering System

- The screen dialog should appear as follows:

```
-----Menu-----
1
2
3
2
0
Total Cost: 140

Process returned 0 (0x0)   execution time : 6.617 s
Press any key to continue.
```


Evaluation

- Lab8.1(30%)
  - 5 testcases, 4% each
- Lab8.2(30%)
  - 5 testcases, 4% each
- Lab8.3(30%)
  - 5 testcases, 4% each
- Lab8.4(20%)
  - 5 testcases, 8% each





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